

MODULE 3

DOCUMENTATION USING MARKDOWN

Imagine opening a project and seeing files—but no explanation.

`project.py data.csv results.png README.md`

Markdown turns plain text into a readable story

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“How do I understand this project?”

Markdown gives ordinary text simple symbols that GitHub can turn into headings, lists, links, code and more.

METAPHOR: Markdown = signboards that guide your reader.

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Headings, Bold & Italics

Start with three simple tools for making information easier to scan.

HEADING

My Project

About

Results

BOLD

****Important****

Make a key word stand out.

ITALICS

Note

Use italics for emphasis.

Challenge: write a title, one bold word and one italic word in a Markdown file.

0	1	Introduction to Markdown
0	2	Headings, Bold & Italics
0	3	Lists, Tables & Checklists
0	4	Links, Images & Emojis
0	5	Code Blocks & Quotes
0	6	README Writing
0	7	Documentation & Reports
0	8	Research & Meeting Notes
0	9	AI-Assisted Documentation
1	0	Final Markdown Challenge

Lists, Tables & Checklists

Use structure when your reader needs to compare, scan or complete tasks.

LISTS

- Python
 - GitHub
 - Markdown
-
1. Create
 2. Edit
 3. Commit

TABLES

Skill Level
--- ---
Python Basic
GitHub Basic

CHECKLISTS

- ☐ Create repo
- ☒ Add README
- ☐ Add image

METAPHOR: lists are shelves; tables are comparison charts; checklists are task boards.

Hyperlinks, Images & Emojis

Documentation becomes more useful when readers can explore and visualize.

HYPERLINK

[Visit
GitHub](https://github.com/)

Take the reader somewhere
useful.

IMAGE

![Project image](image.png)

Use images when they clarify.

EMOJI

 Start
 Complete
 Warning

Use them carefully.

Rule: every visual element should help the reader—not simply decorate the page.

Code Blocks & Quotes

Show technical examples clearly and highlight important statements.

CODE BLOCK

```
``python  
print("Hello GitHub")  
``
```

Fenced code keeps code separate from normal text.

QUOTE

> Documentation is
> written for the reader.

Use quotes for important statements or reminders.

Metaphor: code blocks are glass boxes—readers see the code without confusing it with normal text.

README = the welcome guide

A README should let a first-time visitor understand your project quickly.

A GOOD README ANSWERS

1. What is the project?
2. Why is it useful?
3. How do I start?
4. What technology is used?
5. Who created it?
6. Where can I get help?

Student Attendance Analytics

About
Analyzes attendance data.

Features
- Calculation
- Charts
- Reports

Author
Your Name

METAPHOR: README = a tour guide for someone who has never seen your project.

0 Introduction to
1 Markdown

0 Headings, Bold & Italics
2

0 Lists, Tables & Checklists
3

0 Links, Images & Emojis
4

0 Code Blocks & Quotes
5

0 README Writing
6

0 Documentation &
7 Reports

0 Research & Meeting
8 Notes

0 AI-Assisted
9 Documentation

1 Final Markdown
0 Challenge

Documentation & Reports

Markdown can support structured academic and project writing.

Documentation

Explain how a project works.

Report

Present objectives, methods and results.

User Manual

Explain how to use a system.

Project Notes

Record decisions and information.

The format stays simple; the purpose changes.

0 Introduction to
1 Markdown

0 Headings, Bold & Italics
2

0 Lists, Tables & Checklists
3

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4

0 Code Blocks & Quotes
5

0 README Writing
6

0 Documentation & Reports
7

0 **Research & Meeting**
8 **Notes**

0 AI-Assisted
9 Documentation

1 Final Markdown
0 Challenge

Research Notes & Meeting Notes

Good documentation captures information so another person can understand what happened.

RESEARCH NOTES

Research Question
What are we investigating?

Source
Where did it come from?

Finding
What did we learn?

MEETING NOTES

Date
17 August 2026

Attendees
Team members

Decisions
- Task assigned
- Deadline agreed

Next steps
- Complete draft

Metaphor: documentation is the project's memory for people.

AI can help draft. You remain responsible.

Use AI for structure and clarity—not as a replacement for thinking or verification.

1 Give accurate facts

2 AI creates a draft

3 Check every claim

4 Edit in your own words

5 Verify sources

6 Publish responsibly

Good prompt: “Create a README structure using only the facts I provide.”

Never publish AI-generated claims you have not checked.

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Build a complete Markdown README

Make a stranger understand your project in two minutes.

-  Create a project title
-  Add an About section
-  Add a feature list
-  Add a table
-  Add one hyperlink
-  Add an image or emoji
-  Add a code example
-  Add a checklist
-  Add author information

CREATE → FORMAT → EXPLAIN → VERIFY → PUBLISH

Closing call-to-action: Don't just write documentation. Make your project understandable.

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